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Single Band Motion of Conduction Electrons in a Uniform Magnetic Field

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Abstract. The effect of a uniform magnetic field on the conduction band of metal is investigated, using as model the tight-binding approximation for a simple cubic crystal. The normally discrete magnetic levels pertaining to free electrons are shown to be non-uniformly spaced and broadened as a result of the lattice forces.

§ 1. INTRODUCTION

THE effect of a uniform magnetic field on the conduction electrons in a metal is of general interest for a variety of magnetic phenomena. Little attempt, however, has been made to examine the problem rigorously. For some effects, e.g. diamagnetism, the approximation of an effective mass has been successfully used, thus reducing the problem to that of free electrons, which is then easily handled. This procedure has some support from a theorem first enunciated by Peierls (1933). This states that an approximate Hamiltonian describing the magnetic motion is obtained by replacing $E(\mathbf{k})$ by $E(\mathbf{x})$ where $[\kappa_x, \kappa_y] = (ie/\hbar c)H_z$, etc., H being the magnetic field and $E(\mathbf{k})$ the energy relative to a particular band. This implies immediately that there is no mixing of bands and that they can be treated independently. If, therefore, the occupied levels lie at the top or bottom of a band, $E(\mathbf{k})$ may be represented by the use of effective mass parameters (to describe anisotropy of the metal) and if one takes

$$\hbar\mathbf{x} = \mathbf{p} - (e/c)\mathbf{A}(\mathbf{r}), \quad \mathbf{p} = \hbar\nabla/i,$$

$E(\mathbf{x})$ reduces to a quadratic differential operator whose eigenvalues are easily obtained.

The above theorem was initially obtained for the limiting case of tight binding. A more general derivation has been given by Luttinger (1951) using Wannier functions. There remain, however, several obscurities as regards the interpretation of $E(\mathbf{x})$; more important, no useful estimate of the errors involved is provided. In this paper the theorem is assumed and used to investigate magnetic eigenvalues in the case of tight binding for a simple cubic lattice. This special case serves to illustrate features common to more complicated band structures. What is important to notice is that motion in a particular band is considered independent of other bands, interband terms being omitted. This is usually described as the single-band approximation. A subsequent paper will complete the examination and indicate in what circumstances the approximation is justified (Harper 1955).

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§ 2. SOLUTION OF THE EIGENEQUATION BY A W. K. B. METHOD

The eigenequation considered is

$$E(\mathbf{r})u(\mathbf{r}) = Wu(\mathbf{r}) \quad \dots\dots(1)$$

where

$$\hbar\mathbf{x} = \frac{\hbar}{i}\nabla - \frac{e}{c}\mathbf{A}(\mathbf{r}). \quad \dots\dots(1.1)$$

A discussion of the exact meaning of $u(\mathbf{r})$ is reserved for a subsequent paper. It is sufficient at present to regard it as the actual wave function. Then a convenient gauge for $\mathbf{A}$ is $\mathbf{A} = (0, Hx, 0)$, i.e. $\mathbf{H}$ is directed along the z -axis. In the case of tight binding, $E = \delta(\cos k_x a + \cos k_y a + \cos k_z a)$ if the field $\mathbf{H}$ is assumed to coincide with a crystallographic axis. Then (1) and (1.1) give

$$\begin{aligned} \frac{1}{2}[u(x+a) + u(x-a) + \exp(i\epsilon Hxa)u(y+a) + \exp(-i\epsilon Hxa)u(y-a) \\ + u(z+a) + u(z-a)] = Wu(\mathbf{r}). \end{aligned}$$

$u(\mathbf{r}) \equiv \exp(ik_y y) \exp(ik_z z)u(x)$ separates the variables, giving

$$\frac{1}{2}[u(x+a) + u(x-a)] + \cos(\epsilon a Hx - k_y a)u(x) = (W - \cos k_z a)u(x) \quad \dots\dots(1.2)$$

where $\epsilon \equiv e/\hbar c$, and W is supposed measured in units of δ . The energy due to motion in the field direction is thus additive. Including it in the eigenvalue W , and changing the origin of x , (1.2) may be more simply presented as a difference equation,

$$\frac{1}{2}(u_{n+1} + u_{n-1}) + (\cos \alpha n - W)u_n = 0, \quad \dots\dots(1.3)$$

where $\alpha = \epsilon a^2 H$. For fields $H \sim 10^4$ oersteds, and taking $a \sim 10^{-8}$ cm, $\alpha \sim 10^{-4}$. Consequently $\cos \alpha n$ may be regarded as slowly varying, and approximate eigenvalues may be found using an adaptation of the W.K.B. method. For convenience write $\cos \alpha n - W$ as k_n . Then for k_n less than 1, u_n is oscillatory; for k_n greater than 1, u_n behaves exponentially. $k_n = 1$ is therefore a classical turning point and in its vicinity u_n clearly behaves linearly.

It is now evident that $|W| < 2$, since for $|W|$ greater than 2, $|k_n|$ is always greater than 1 and therefore boundary conditions cannot be satisfied. Thus the eigenvalues W are confined to a band whose energy limits are identical with those occurring in the non-magnetic case. The levels are, of course, discrete (apart from the field-direction energy $\cos k_z a$) and may be obtained as follows:

For $|k_n| < 1$, put $u_n = \exp i\phi_n$. Then (1.3) becomes

$$\exp(i\Delta\phi_{n+1}) + \exp(-i\Delta\phi_n) + 2k_n = 0$$

where $\Delta\phi_n$ denotes the first difference $\phi_n - \phi_{n-1}$, i.e.

$$\exp\{i(\Delta\phi_n + \Delta^2\phi_{n+1})\} + \exp(-i\Delta\phi_n) + 2k_n = 0.$$

Assuming now that $\Delta\phi_n$ is slowly varying, then $\Delta^2\phi_{n+1} \sim \Delta^2\phi_n \ll 1$, and therefore,

$$\cos \Delta\phi_n + \frac{1}{2}i \exp(i\Delta\phi_n) \Delta^2\phi_n \simeq -k_n.$$

As a zero-order approximation, $\cos \Delta\phi_n^0 = -k_n$. Put $\Delta\phi_n = \Delta\phi_n^0 + \chi_n$, χ_n being of the same order of magnitude as $\Delta^2\phi_n$, then

$$\cos \Delta\phi_n^0 - \chi_n \sin \Delta\phi_n^0 + \frac{1}{2}i \Delta^2\phi_n^0 \exp(i\Delta\phi_n^0) \simeq -k_n,$$

or

$$\chi_n = \frac{1}{2}i(\cot \Delta\phi_n^0 + i)\Delta^2\phi_n^0.$$

Consequently,
$$\phi_n = \sum^n (\Delta\phi_l^0 + \chi_l)$$

$$= \sum^n \Delta\phi_l^0 + \frac{1}{2}i \log \sin \Delta\phi_n^0 - \frac{1}{2}\Delta\phi_n^0 + \text{const.}$$

(replacing $\sum_l^n$ by $\int^n dl$).

Thus

$$u_n \simeq \text{const.} (\sin \Delta\phi_n^0)^{-1/2} \exp \left(\pm i \int^n \Delta\phi_l^0 dl \right). \quad \dots\dots (1.4)$$

This is in obvious analogy to the usual results for a continuous variable. For $k_n > 1$, the corresponding expression is

$$u_n \simeq \text{const.} (\sinh \Delta\phi_n^0) \exp \pm \int^n \Delta\phi_l^0 dl, \quad \dots\dots (1.5)$$

where $\cosh \Delta\phi_n^0 = -k_n$. These approximations fail near the turning point $k_n = 1$, since then $\Delta^2\phi_n^0$ is not small but of the order of unity.

In the usual way suitable combinations of these solutions have to be matched in order to provide eigenvalues. In the region $k_n \sim 1$, $\cos \alpha n - W$ may be expanded to $1 - \alpha(n - n_0) \sin \alpha n_0$ where $\cos \alpha n_0 - W = 1$. k_n is therefore linear in n and two independent solutions are

$$J_{n-n_0-m} \left(\frac{2}{m} \right), \quad Y_{n-n_0-m} \left(\frac{2}{m} \right) \quad \dots\dots (1.6)$$

where $m = 1/\alpha \sin n_0 \alpha$. For $|m| \gg 1$, and $n \sim n_0$, both these expressions behave linearly (see Nicholson's formulae, Watson 1944, p. 249) in accordance with the earlier remark. To effect matching, asymptotic expansions are available (Watson 1944, p. 244) and the arguments run parallel to those usually presented for the case of a continuous variable.

There is one important difference, namely that the turning points n_0 are infinite in number, and cyclic boundary conditions are necessary. It is clear that $u_{n+2\pi/\alpha}$ satisfies the same equation as u_n . It is permissible therefore to choose solutions u_n such that

$$u_{n+2\pi/\alpha} = u_n \exp(2\pi i k_x a / \alpha)$$

i.e.

$$u_n = v_n \exp(i k_x n a), \quad -\frac{1}{2}\alpha < k_x a < \frac{1}{2}\alpha$$

where now v_n is periodic with period $2\pi/\alpha$. Equation (1.3) thus becomes

$$v_{n+1} \exp(i k_x a) + v_{n-1} \exp(-i k_x a) + v_n (\cos \alpha n - W) = 0. \quad \dots\dots (1.7)$$

This simple procedure is, strictly speaking, correct only for special values of α , namely if $2\pi/\alpha$ is integral, or $N = 2\pi/\epsilon a^2 H$ an integer. This, however, is a mild restriction on the field since $N \sim 10^5$ for $H \sim 10^4$ oersteds. It is clear that only positive W (say) need be considered since putting $v_n = (-1)^n \omega_n$ and changing the origin of n by π/α reverses the sign of W . The eigenvalues are therefore symmetrical about $W = 0$.

The effect of phase factors $\exp(\pm i k_x a)$ in (1.7) is to add $k_x a$ to $\Delta\phi_n$ throughout the analysis. Thus $\Delta\phi_n^0$ is determined by

$$\cos(k_x a + \Delta\phi_n^0) + \cos \alpha n - W = 0. \quad \dots\dots (1.8)$$

The matching of (1.4), (1.5) is carried out using the linear solutions (1.6) and their asymptotic expansions. The phase of (1.4) is determined and as usual is corrected by $\pi/4$. From familiar arguments quantization is then effected by requiring that

$$\int_{n_0}^{2\pi/\alpha - n_0} \Delta\phi_l^0 dl = (j + \frac{1}{2})\pi \quad \dots\dots(1.9)$$

the upper and lower limits of the integral being two adjacent turning points which contain the oscillatory part of u_n .

§ 3. LEVEL BROADENING

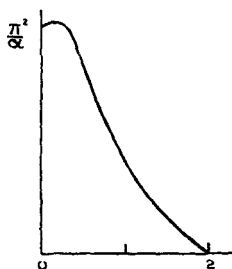
Equations (1.8) and (1.9) are a particular instance of Onsager's results (1952). For completeness one should also take account of the phase $k_y a$ occurring in $\cos \alpha n$. There are then two parameters k_x, k_y , in addition to j , necessary to describe completely the level spectrum. It will presently appear that the former describe a broadening of a (nearly) discrete level j .

Omitting k_x, k_y for the present, the equation for eigenvalues is

$$\int_{-n_0}^{+n_0} \cos^{-1}(W - \cos \alpha n) dn = (j + \frac{1}{2})\pi \quad \dots\dots(1.10)$$

where $\cos \alpha n_0 - W + 1 = 0$, $0 < \alpha n_0 \leq \pi$.

Plotting the integral in (1.10) as a function of W (see figure), one sees for example, that the unbroadened levels become more widely separated near $W=0$.



To examine broadening, consider k_x only. Then

$$\Delta\phi_n = -k_x \alpha + \cos^{-1}(W - \cos \alpha n),$$

and (1.9) becomes

$$\int_{-n_0}^{+n_0} \cos^{-1}(W - \cos \alpha n) dn - 2n_0 k_x a = (j + \frac{1}{2})\pi$$

or

$$\int_{-n_0}^{+n_0} \cos^{-1}(W - \cos \alpha n) dn = (j + \frac{1}{2})\pi + \frac{2k_x a}{\alpha} \cos^{-1}(W - 1). \quad \dots\dots(1.11)$$

Since $-\frac{1}{2}\alpha < k_x a < \frac{1}{2}\alpha$ the broadening term is always less than π so that the levels are always separated. Using equation (1.1) then the levels may be described as follows. For $W=2-\omega$, $\omega \ll 1$, an approximate solution is

$$\omega = (j + \frac{1}{2})\alpha + 2k_x a [(2j + 1)a]^{1/2}.$$

Thus at the edge of the band the levels are uniformly spaced, the separation being given by the effective mass in the customary way. There is in addition a broadening, but so slight as to be negligible. Qualitatively the figure indicates

how the levels vary throughout the band, the most important fact being that near the centre, $W=0$, the broadening is extremely large, the actual separation in fact being small compared with it. The situation here then is just the inverse of that occurring at the edge of the band.

It should be remarked that, strictly speaking, the above equations do not hold at the very centre $W=0$, since for this value the turning points n_0 are no longer linear and the preceding analysis does not apply. On grounds of continuity, however, the conclusions seem fairly assured.

Concerning Onsager's work it is perhaps worth emphasizing that the magnetic quantization of levels is a direct consequence of $[\kappa_x, \kappa_y] = i e H_z$ etc. and that the notion of wave packets is unnecessary.

REFERENCES

- HARPER, P. G., 1955, *Proc. Phys. Soc. A*, **68**, 879.
 LUTTINGER, J. M., 1951, *Phys. Rev.*, **84**, 814.
 ONSAGER, L., 1952, *Phil. Mag.*, **43**, 1006.
 PEIERLS, R. E., 1933, *Z. Phys.*, **80**, 763.
 WATSON, G. N., 1944, *A Treatise on the Theory of Bessel Functions*, 2nd Edn (Cambridge University Press).